



strace

System call tracer - <https://strace.io>

- ▶ Available on all GNU/Linux systems
Can be built by your cross-compiling toolchain generator or by your build system.
- ▶ Allows to see what any of your processes is doing: accessing files, allocating memory... Often sufficient to find simple bugs.
- ▶ Usage: `strace <command>` (starting a new process) `strace -f <command>` (**f**ollow child processes too) `strace -p <pid>` (tracing an existing process) `strace -c <command>` (time statistics per system call) `strace -e`

`<expr> <command>` (use **e**xpression for advanced filtering)

See [the strace manual](#) for details



strace



Image credits: <https://strace.io/>



strace example output

```
> strace cat Makefile
[...]  
fstat64(3, {st_mode=S_IFREG|0644, st_size=111585, ...}) = 0  
mmap2(NULL, 111585, PROT_READ, MAP_PRIVATE, 3, 0) = 0xb7f69000  
close(3) = 0  
access("/etc/ld.so.nohwcap", F_OK) = -1 ENOENT (No such file or directory)  
open("/lib/tls/i686/cmov/libc.so.6", O_RDONLY) = 3  
read(3, "\177ELF\1\1\1\0\0\0\0\0\0\0\3\0\3\0\1\0\0\0\320h\1\0004\0\0\0\344"... , 512) = 512  
fstat64(3, {st_mode=S_IFREG|0755, st_size=1442180, ...}) = 0  
mmap2(NULL, 1451632, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0xb7e06000  
mprotect(0xb7f62000, 4096, PROT_NONE) = 0  
mmap2(0xb7f66000, 9840, PROT_READ|PROT_WRITE,  
      MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0xb7f66000  
close(3) = 0  
[...]  
openat(AT_FDCWD, "Makefile", O_RDONLY) = 3  
newfstatat(3, "", {st_mode=S_IFREG|0644, st_size=173, ...}, AT_EMPTY_PATH) = 0  
fadvise64(3, 0, 0, POSIX_FADV_SEQUENTIAL) = 0  
mmap(NULL, 139264, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f7290d28000  
read(3, "ifneq ($(KERNELRELEASE),)\nobj-m "... , 131072) = 173  
write(1, "ifneq ($(KERNELRELEASE),)\nobj-m "... , 173ifneq ($(KERNELRELEASE),)
```

Hint: follow the open file descriptors returned by `open()`. This tells you what files are handled by further system calls.



strace filtering

- ▶ Display only a specific set of system calls:

```
$ strace -e 'openat,write' cat Makefile
```

- ▶ Filter out specific system calls:

```
$ strace -e '!poll' cat Makefile
```

- ▶ Show only system calls returning a specific status

```
$ strace -e 'status=failed' cat Makefile
```

- ▶ Trace how a file is accessed and used among different system calls

```
$ strace -P '/etc/ld.so.cache' cat Makefile
```

- ▶ Run `strace -tips` to learn new commands !